SPESIFIKASI TEKNIS & DOKUMENTASI DESAIN SISTEM (DOCSHEET)

Nama Proyek: ALUR Logistik (*LajuLogistik*)
Versi Dokumen: 1.0.0
Tanggal: September 2026
Status: Approved for Development

1. Ringkasan Sistem & Arsitektur

ALUR Logistik adalah sistem manajemen pengiriman dan pelacakan paket berbasis status *real-time*. Sistem ini membagi antarmuka menjadi dua kategori: **Progressive Web App (PWA)** berbasis *mobile-first* untuk Customer dan Kurir, serta **Desktop Web Dashboard** untuk Admin.

Tech Stack Utama

- **Frontend Customer & Kurir:** Vue 3, Vite, Tailwind CSS, vite-plugin-pwa (PWA).
- **Frontend Admin Dashboard:** Vue 3, PrimeVue / Tailwind CSS (Full Desktop UI).
- **Backend API:** FastAPI (Python 3.11+), SQLAlchemy 2.0 Async, Pydantic v2.
- **Database:** PostgreSQL 13+.
- **Background Task / Scheduler:** APScheduler (Running inside FastAPI async event loop).
- **Authentication:** OAuth2 dengan JSON Web Tokens (JWT) & Role-Based Access Control (RBAC).

2. Matriks Peran & Hak Akses (RBAC)

Modul / Fitur	Customer (PWA)	Courier (PWA)	Admin (Desktop)
Self-Registration Account	**Ya**	Tidak	Tidak
Create Delivery Request (DR)	**Ya**	Tidak	Tidak
View Own Tracking History	**Ya**	Tidak	Tidak
Confirm Completed Order	**Ya**	Tidak	Tidak
View Assigned Task List	Tidak	**Ya**	Tidak

Update Status (Picked Up, In Transit, Delivered)	Tidak	**Ya**	Tidak
Upload Proof of Delivery Image	Tidak	**Ya**	Tidak
View All Shipments & Filter	Tidak	Tidak	**Ya**
Assign Courier to DR	Tidak	Tidak	**Ya**
Reject DR (+ Input Alasan)	Tidak	Tidak	**Ya**
Courier Account CRUD	Tidak	Tidak	**Ya**
Manage System Settings	Tidak	Tidak	**Ya**

3. Aturan Bisnis & Lifecycle Status

3.1 Status Order Lifecycle





3.2 Logika Khusus & Triggers

1. **Visual Alert (Highlight Baris Merah di Admin UI):**
   - **Pemicu:** Status shipment = PENDING.
   - **Kondisi:** $\text{Waktu Sekarang} - \text{created_at} \ge \text{pending_alert_timeout_minutes}$ (Default: 30 Menit).
   - **Efek UI:** Baris tabel di Admin Dashboard berubah latar belakang menjadi **MERAH** dan posisinya diprioritaskan ke paling atas (*sorting priority*).
2. **Auto-Complete (Cron Job Background Task):**
   - **Pemicu:** Task berjalan setiap 1 jam.
   - **Kondisi Query:** Status = DELIVERED DAN $\text{delivered_at} \le \text{NOW()} - \text{INTERVAL '3 days'}$.
   - **Efek Sistem:** Status otomatis berubah menjadi COMPLETED dengan record tracking_histories.created_by = NULL.

4. Kamus Data Database (Data Dictionary)

4.1 Tabel users

Menyimpan seluruh data pengguna sistem (Customer, Courier, Admin).

Nama Kolom	Tipe Data	Constraint	Deskripsi
id	BIGSERIAL	PRIMARY KEY	Identitas unik user
name	VARCHAR(100)	NOT NULL	Nama lengkap user
email	VARCHAR(150)	UNIQUE, NOT NULL	Email untuk login
phone	VARCHAR(20)	UNIQUE, NOT NULL	Nomor telepon / WhatsApp
password_hash	VARCHAR(255)	NOT NULL	Hash kata sandi (Bcrypt)
role	ENUM	NOT NULL	'ADMIN',

			'COURIER', 'CUSTOMER'
is_active	BOOLEAN	DEFAULT TRUE	Status keaktifan akun
created_at	TIMESTAMPTZ	DEFAULT NOW()	Waktu registrasi
updated_at	TIMESTAMPTZ	DEFAULT NOW()	Waktu pembaruan akun

4.2 Tabel shipments

Tabel utama menyimpan data permintaan pengiriman dan status terkini.

Nama Kolom	**Tipe Data**	**Constraint**	**Deskripsi**
id	BIGSERIAL	PRIMARY KEY	Identitas unik shipment
tracking_number	VARCHAR(50)	UNIQUE, NOT NULL	Nomor resi (TRX-YYYYMMDD-XXXX)
customer_id	BIGINT	FK -> users.id	Pembuat request pengiriman
courier_id	BIGINT	FK -> users.id, NULLABLE	Kurir yang ditugaskan
sender_name	VARCHAR(100)	NOT NULL	Nama pengirim
sender_phone	VARCHAR(20)	NOT NULL	Nomor telepon pengirim
origin_address	TEXT	NOT NULL	Alamat lengkap penjemputan

recipient_name	VARCHAR(100)	NOT NULL	Nama penerima
recipient_phone	VARCHAR(20)	NOT NULL	Nomor telepon penerima
destination_address	TEXT	NOT NULL	Alamat lengkap tujuan
package_description	TEXT	NOT NULL	Deskripsi isi paket
weight_kg	NUMERIC(6,2)	NOT NULL	Berat barang (kg)
status	ENUM	DEFAULT 'PENDING'	'PENDING', 'REJECTED', 'ASSIGNED', 'PICKED_UP', 'IN_TRANSIT', 'DELIVERED', 'COMPLETED'
rejection_reason	TEXT	NULLABLE	Catatan jika di-reject Admin
proof_image_url	VARCHAR(255)	NULLABLE	URL foto bukti pengiriman
delivered_at	TIMESTAMPTZ	NULLABLE	Timestamp saat status DELIVERED
created_at	TIMESTAMPTZ	DEFAULT NOW()	Waktu pengajuan DR
updated_at	TIMESTAMPTZ	DEFAULT NOW()	Waktu pembaruan status terakhir

4.3 Tabel tracking_histories

Pencatatan riwayat audit trail setiap perubahan status paket.

Nama Kolom	Tipe Data	Constraint	Deskripsi
id	BIGSERIAL	PRIMARY KEY	Identitas unik log
shipment_id	BIGINT	FK -> shipments.id	ID shipment terkait
status	ENUM	NOT NULL	Status pada tahap tersebut
notes	TEXT	NOT NULL	Keterangan/catatan log
created_by	BIGINT	FK -> users.id, NULLABLE	ID Aktor (NULL jika System/Cron)
created_at	TIMESTAMPTZ	DEFAULT NOW()	Waktu log dicatat

4.4 Tabel system_settings

Konfigurasi variabel dinamis sistem.

Nama Kolom	Tipe Data	Constraint	Deskripsi
key	VARCHAR(50)	PRIMARY KEY	Kunci konfigurasi
value	VARCHAR(255)	NOT NULL	Nilai konfigurasi
description	TEXT	NULLABLE	Deskripsi parameter
updated_at	TIMESTAMPTZ	DEFAULT NOW()	Waktu pembaruan

5. Kontrak REST API (API Specifications)

5.1 Authentication (/api/v1/auth)

HTTP
POST /api/v1/auth/register
Content-Type: application/json
Access: Public (Customer Only)

Payload:

```
{
  "name": "Eko Customer",
  "email": "eko@gmail.com",
  "phone": "081234567892",
  "password": "Password123!"
}
```

Response (201 Created):

```
{
  "message": "Registrasi berhasil",
  "user_id": 3
}
```

HTTP
POST /api/v1/auth/login
Content-Type: application/json
Access: Public

Payload:

```
{
  "email": "admin@alur.com",
  "password": "Password123!"
}
```

Response (200 OK):

```
{
  "access_token": "eyJhbGciOiJKV1QiLC...",
  "token_type": "bearer",
  "role": "ADMIN",
  "name": "Super Admin"
}
```

5.2 Shipments Core (/api/v1/shipments)

```
HTTP
POST /api/v1/shipments
Authorization: Bearer <token>
Access: Customer

Payload
{
  "sender_name": "Eko Customer"
  "sender_phone": "081234567892"
  "origin_address": "Jl. Kemang Raya No. 10, Jakarta Selatan"
  "recipient_name": "Budi Santoso"
  "recipient_phone": "089876543210"
  "destination_address": "Jl. Asia Afrika No. 5, Bandung"
  "package_description": "Dokumen Kontrak Kerja"
  "weight_kg": 0.5
}

Response (201 Created)
{
  "id": 101
  "tracking_number": "TRX-20260925-0001"
  "status": "PENDING"
  "created_at": "2026-09-25T10:00:00Z"
}
```

```
HTTP
GET /api/v1/shipments/{tracking_number}/track
Access: Public / Authenticated

Response (200 OK)
{
  "tracking_number": "TRX-20260925-0001"
  "status": "IN_TRANSIT",
```

```
  "sender_name": "Eko Customer",
  "recipient_name": "Budi Santoso",
  "destination_address": "Jl. Asia Afrika No. 5, Bandung",
  "history": [
    {
      "status": "PENDING",
      "notes": "Delivery request berhasil dibuat oleh Customer.",
      "created_at": "2026-09-25T10:00:00Z"
    },
    {
      "status": "ASSIGNED",
      "notes": "Pesanan telah disetujui Admin dan ditugaskan ke Kurir Budi.",
      "created_at": "2026-09-25T10:15:00Z"
    },
    {
      "status": "PICKED_UP",
      "notes": "Paket telah diambil oleh kurir dari lokasi pengirim.",
      "created_at": "2026-09-25T11:00:00Z"
    }
  ]
}
```


5.3 Courier Actions (/api/v1/courier)

```
HTTP
PUT /api/v1/courier/shipments/{id}/status
Authorization: Bearer <token>
Access: Courier

Payload (Update Status biasa)
{
  "status": "IN_TRANSIT",
  "notes": "Paket dalam perjalanan via jalur darat"
}

Payload (Saat Status DELIVERED)
{
  "status": "DELIVERED"
  "notes": "Diterima oleh Ybs"
```

```
  "proof_image_url": "https://storage.alur.com/proofs/trx-0001.jpg"
}
```

```
Response (200 OK)
{
  "message": "Status berhasil diperbarui"
  "current_status": "DELIVERED"
}
```

5.4 Admin Management (/api/v1/admin)

```
HTTP
PUT /api/v1/admin/shipments/{id}/assign
Authorization: Bearer <token>
Access: Admin
```

```
Payload
{
  "courier_id": 2
}
```

```
Response (200 OK)
{
  "message":"Kurir berhasil ditugaskan"
  "shipment_id": 101
  "status": "ASSIGNED"
}
```

```
HTTP
PUT /api/v1/admin/shipments/{id}/reject
Authorization: Bearer <token>
Access: Admin
```

```
Payload
{
```

```
  "rejection_reason": "Alamat pengirim tidak terjangkau layanan operasional."
}
```

```
Response (200 OK)
{
  "message": "Delivery request ditolak",
  "shipment_id": 101,
  "status": "REJECTED"
}
```

```
HTTP
PUT /api/v1/admin/settings
Authorization: Bearer <token>
Access: Admin
```

```
Payload
{
  "pending_alert_timeout_minutes": "45",
  "auto_complete_days": "3"
}
```

```
Response (200 OK)
{
  "message": "Pengaturan sistem berhasil disimpan"
}
```

6. Persyaratan Non-Fungsional (Non-Functional Requirements)

1. **Performa & Latensi API:** Response time untuk REST API standard $\le 200\text{ ms}$ pada kondisi beban normal.
2. **PWA Mobile Capability:**
   - Aplikasi PWA Customer & Kurir dapat diinstal langsung via browser (*Add to Home Screen*).
   - Mendukung Web Push Notification untuk pembaruan status order.
   - Integrasi kamera native HP untuk fungsi *Upload Proof of Delivery*.
3. **Keamanan Data:**
   - Seluruh komunikasi wajib menggunakan enkripsi **HTTPS/TLS 1.3**.

- Password hashing menggunakan standar **Bcrypt** dengan *cost factor* 12.
- Akses API dilindungi dengan token **JWT Bearer** yang memiliki masa kadaluarsa (misal: 24 jam).

4. **Ketersediaan Scheduler (Cron Job):** Background task harus memiliki mekanisme *retry logic* jika koneksi database terputus saat mengeksekusi *auto-complete*.